When Algorithms Err: Differential Impact of Early vs. Late Errors on Users' Reliance on Algorithms

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Abstract

Errors are a natural part of the development and use of predictive algorithms, but they could discourage people from relying on algorithms even when doing so could lead to better decisions. In this paper, we conduct two experiments to demonstrate that people's reliance on a predictive algorithm following a substantial error depends on when the error occurs and how the algorithm is used in decision making. We find that, when the prediction tasks are fully delegated to an algorithm, the impact of an error on reliance is different if the error occurs *early* versus *late*. While an early error results in substantial and persistent reliance reduction, a late error affects reliance only temporarily. However, when users have more control over how to use the algorithm's predictions, the risk associated with early errors decreases. Our work advances the understanding of algorithm aversion and informs the practical design of algorithmic decision-making systems.

Keywords: Algorithm, decision support, prediction, timing of error, reliance.

1 Introduction

The abundance of data, exponential growth in computational capability, and advances in predictive algorithms and models have all culminated in the age of artificial intelligence (AI) and analytics. Across a number of different domains, predictive algorithms have been shown to outperform human judgments (e.g., Schweitzer and Cachon, 2000; Highhouse, 2008). Even in areas that were once considered to be dominated by human intelligence, we are witnessing

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examples of algorithm superiority. For example, the field of AI celebrated a huge breakthrough in 2016 when Google's AI beat a top player at the game of Go, which is known to be exponentially more complex than chess (Metz, 2016). Now, autonomous vehicles are considered safer than human drivers as they are equipped with better sensors and are not affected by substances or fatigue (Walker, 2020).

In many situations, however, humans are still in the "driver's seat" and choose whether or not to use algorithms for decision making. Despite empirical evidence of algorithm's superiority, decision-makers are often averse to using algorithms, relying instead on their own (often less accurate) judgments (Fildes and Goodwin, 2007; Vrieze and Grove, 2009). Even in cases where people do rely on algorithms, they tend not to be forgiving when algorithms make an error, revealing that the algorithms may not be perfect (Dietvorst et al., 2015). People's reluctance to use superior but imperfect algorithms in favor of inferior human judgment poses a major challenge for organizations aiming to make more accurate and informed decisions as well as for firms that develop and sell algorithmic tools.

Because real-world situations are often far from perfectly predictable, even the best predictive algorithms produce imperfect predictions. While it is known that errors lead to *algorithm aversion* (Dietvorst et al., 2015), in this paper, we posit that not all errors affect users' reliance on the algorithm to the same extent.[1] In particular, we focus on the timing of algorithmic errors and compare the differential impacts of *early* errors (i.e., errors that occur when users first start to use the algorithm) with *late* errors (i.e., errors that occur after users have used the algorithm for an extended period) on users' reliance on the algorithm. Studying the timing of algorithmic errors is not only interesting from a theoretical perspective but also relevant to practice. As an example, algorithmic systems sometimes need to choose between making "safer" decisions that have worked well in the past and exploring "riskier" decisions that might potentially lead to substantial errors. In this context, the timing of errors and their differential impacts on reliance are important factors to consider. Addition-

[1]Note that an "error" in our paper refers to a substantial mistake of the algorithm that is worse than its expected performance.

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ally, understanding the impact of error timing can also inform the decisions regarding how much to train and test an algorithmic tool before offering it to the end-users. Balancing the cost of training and the risk of errors is an integral part of the development and use of algorithms.

We conducted two experimental studies on the Amazon Mechanical Turk platform and asked participants to perform a series of prediction tasks. We manipulated a predictive algorithm to make a substantial error at either the early stage or the late stage of the experiment. In the first study, participants chose to either completely rely on the algorithm or rely on their own judgments for each prediction task. We found that, for participants exposed to an early error, their overall reliance on the algorithm experienced a large drop and never quite recovered to the baseline level (i.e., reliance level in the control group where no substantial error occurred) within the duration of the entire experiment. However, participants exposed to the late error reacted differently. While their reliance dipped immediately after the error, it quickly recovered in subsequent rounds and showed no difference from the baseline level. In other words, early errors can lead to *persistent* algorithm aversion, which is more detrimental to how people perceive and use predictive algorithms compared to late errors.

In the second study, participants had more control in using the algorithm. Instead of completely delegating the task to the algorithm, the participants were shown the algorithm's prediction first and then offered an opportunity to incorporate that information to form their final prediction. In this setting, we observed that the reliance was minimally affected even after encountering a substantial error in the early stage. Overall, early errors contribute more to persistent algorithm aversion compared to late errors, but, when can be afforded, giving users greater control can mitigate the risk associated with early errors.

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2 Relevant Literature

2.1 Algorithm Reliance

The nascent literature on algorithm reliance offers several insights regarding when users rely on algorithms for decision making. In one set of studies, researchers have found that people tend to be less tolerant of errors made by algorithms than those by humans, and avoid using algorithms in decision making after seeing the algorithms make mistakes (Dietvorst et al., 2015; Prahl and Van Swol, 2017; Ireland, 2019). Such "algorithm aversion" persists even if the imperfect algorithms still outperform human decision-makers (Dietvorst et al., 2015). Meanwhile, other studies found "algorithm appreciation." For example, Logg et al. (2019) showed that people are more likely to adopt recommendations from algorithms than those from human experts in forecasting tasks. However, this preference for algorithms decreases when people make choices to rely on the algorithms versus their *own* judgments. In general, the degree of algorithm reliance depends on 1) the performance of the algorithm, 2) the nature of decision tasks, and 3) how the algorithm is used.

First, both the stated and the observed performance of an algorithm can affect people's reliance on it (Yin et al., 2019). The impact of algorithm performance on reliance is found to be asymmetric (Yu et al., 2017). Specifically, a substantial decline in trust is observed after a few consecutive system failures, but only a modest increase in trust is shown following consecutive system successes. Second, the nature of the decision task plays an important role. People are more likely to trust and use algorithms in tasks with high perceived objectivity (e.g., logical, rule-based analyses) (Castelo et al., 2019), but instead rely on human judgment for subjective tasks (Yeomans et al., 2019). Third, algorithm reliance also depends on how the algorithm's outputs can be used by decision makers. For example, Dietvorst et al. (2018) found that people are more reliant on the algorithm than their own forecasts if they can modify the algorithm's predictions, even by a very small amount. More generally, having humans "in the loop" by giving them autonomy to incorporate, modify, or override

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algorithms' recommendations can improve reliance (De-Arteaga et al., 2020).

2.2 Reliance Recovery following Algorithm Errors

Practical applications of algorithms often involve repeated interactions with human users and users' reliance on an algorithm can evolve over time. What factors affect the recovery of reliance following algorithm errors? This question is underexplored in the literature on algorithm reliance. However, studies about interpersonal trust can provide some relevant insights on this issue.

Research in interpersonal trust breach and recovery suggests that trust recovery depends on *when* the breach takes place. Schilke et al. (2013) conducted experiments in which participants engaged in repeated trust-honor games with a computer program (framed by the experimenters as other human participants). In each round of the game, a participant decided whether to keep a certain amount of money to herself or to transfer it to the program, who subsequently chose whether to reciprocate or to keep all of the money (thereby violating the participant's trust). The experimental results showed that trust recovered to a higher level following a late breach than an early breach. In other words, a long relationship history facilitated the recovery of trust following a breach.

The literature on impression formation provides another lens to make sense of the relationship between the timing of trust breach and trust recovery. First impression plays a critical role in interpersonal relationships such that the first few pieces of information weigh disproportionally more in impression formation than later ones (Asch, 1946). This is often referred to as "first impression bias" (Lim et al., 2000). Therefore, a trust breach in an early stage of a relationship can leave a negative impression that is hard to overcome, whereas a breach later on is more likely to be forgiven.

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2.3 Enhancing Reliance on Algorithms

Understanding of the factors that affect algorithm reliance also gives rise to potential mechanisms to enhance reliance. Notably, allowing users to modify the algorithm's outputs, thereby granting them control over the decision outcomes, has proven to be an effective strategy to mitigate algorithm aversion (Dietvorst et al., 2018). This is particularly relevant to our study, as it represents a general-purpose mechanism that can be applied to various decision tasks in different domains.

When a human user decides whether to fully delegate a decision task to the algorithm or use her own judgment, reliance on the algorithm is binary; the user either completely relies on the algorithm or not at all. In contrast, when the algorithm's outputs can be incorporated by the user into the decision process to reach a final decision, the user retains control over the decision, and the reliance on the algorithm manifests as the degree to which the user leverages the algorithm's outputs. In other words, whether or not humans have the control over the final decision when using the algorithm determines the nature of reliance on the algorithm, and may affect how reliance changes as a result of algorithm error.

The literature on control and delegation suggests that being able to control one's own decisions and choices is regarded as necessary to maintain individual autonomy (Ryan and Deci, 2000). For example, the autonomy to modify an algorithm's outputs grants users more *control* over the decision outcomes. Even though the algorithm inevitably errs, users are not bound to its errors and can make adjustments to reduce the errors associated with the final decisions (Dietvorst et al., 2018). Additionally, studies of human-robot interactions have found that users attribute less blame to robots for errors made in tasks, if the robots are less autonomous and are directly controlled by users (Kim and Hinds, 2006).

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3 Study 1: Effect of Error Timing on Algorithm Reliance

In Study 1, we manipulated a predictive algorithm to have a substantial prediction error at different stages in a series of prediction tasks and examined how the timing of the substantial error affected an individual's choice to rely on the algorithm in decision making.

3.1 Experiment Design

Our experiment design was motivated by Dietvorst and Bharti (2019). We created a fictitious decision task to predict the yield of a crop field based on three inputs: the amount of rainfall, fertilizer, and sunshine. Each of the three inputs took an integer value between 1 and 25. Unbeknown to the experiment participants, the actual yield was generated based on the following relationship: $Yield = \frac{x+2y+3z}{2} - \frac{x^2+3y^2+4z^2}{80}$, where x, y, z are the amount of rainfall, fertilizer, and sunshine, respectively. This particular functional form was chosen to simulate a hard prediction problem for regular individuals, where the inputs were related to the output in a nonlinear manner, and the specific coefficients were picked so that the yield also largely fell in the same range of 1-25. Each of the three inputs x, y, z was randomly and independently drawn from $\{1, \ldots, 25\}$, and the calculated $Yield$ was rounded to the nearest integer and treated as the true yield of the field. Based on the true yield, we generated the prediction of a fictitious predictive algorithm as $Prediction = Yield + \varepsilon$, with ε randomly drawn from $\{-2, -1, 0, 1, 2\}$. In other words, the algorithm's prediction was unbiased on average and had a deviation no larger than 2 units.

Upon joining the experiment, a participant was informed that she needed to complete 20 rounds of the yield prediction task and that her payment from the experiment would be based on the accuracy of her predictions. During each prediction task, the participant viewed the amount of rainfall, fertilizer, and sunshine, and made an initial prediction of crop yield. Then, she was asked to choose whether to use her initial prediction or rely on

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the algorithm's prediction for the task's final prediction.[2] Upon completion of each task, the participant saw a performance summary page that included the true corp yield in this round, the algorithm's prediction, the error of the participant's final prediction, and the participant's mean absolute prediction error so far. Importantly, the algorithm's prediction for each task was only displayed on the performance summary page, i.e., it was not visible to the participant when she decided to rely on herself or the algorithm. Therefore, reliance on the algorithm represents a full delegation of the decision to the algorithm.[3]

As part of the instruction, a table of 10 records, including the three input values, the corresponding corp yields, and the algorithm's predictions, was displayed to participants. This table offered participants a preliminary understanding of the prediction task and the general quality of the algorithm. Participants also completed a practice task before proceeding to the 20 rounds of actual tasks.

We employed a between-subjects design. Participants were randomly assigned to one of three experimental groups. In the *control* group, participants completed 20 yield predictions, and the algorithm's predictions were always within 2 units of the ground truth. In the *early-error* treatment group, we manipulated the algorithm's prediction in the 2nd round to be 10 units away from the ground truth. Given the algorithm's typical prediction error of no more than 2 units, the 10-unit deviation represented a *substantial* error. Specifically, if the true yield was lower than 15, the algorithm's prediction was $Yield + 10$; otherwise, it was $Yield - 10$. In the *late-error* treatment group, we manipulated the algorithm's prediction error in the 12th round to be 10 units. We chose the 12th round in order to give participants in this group enough time to become familiar with the algorithm and build confidence in it.[4] Note that, for each treatment group, the manipulation took place at the end of the

[2]For a given task, we refer to a participant's prediction based on input data as the "initial prediction". The participant's "final prediction" equals the algorithm's prediction if she chose to rely on the algorithm for the task; otherwise, it equals her initial prediction.

[3]One could also consider a case where users decide whether to rely on the algorithm or not after seeing its prediction. In such a design, users can usually incorporate the algorithm's prediction in their final decisions. Our Study 2 explores this scenario.

[4]A post-hoc analysis showed that participants in the control group were significantly more likely to rely on algorithm's predictions after 10 rounds than they were at the beginning of the experiment. This lends

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treatment round, i.e., the substantial error was revealed only on the performance summary page, after a participant already made her choice of relying on the algorithm or on her initial prediction. Regardless of whether a participant chose to rely on the algorithm's prediction in the treatment round, she always observed the substantial error. The order of the 20 prediction tasks was randomized for each participant.

After completing the 20 prediction tasks, the participants were asked to fill out a questionnaire that collected demographic information about them. They also reported the overall confidence in their own predictions using a 5-point Likert scale. Additionally, they were asked to estimate the mean absolute error (MAE) of their own predictions. Lastly, to assess the participant's tendency of trusting and adopting predictive algorithms and information technologies in general, the questionnaire included two sets of measurement instruments: (1) a 3-item scale for *trust*, adapted from Pavlou and Dimoka (2006), and (2) a 4-item scale for *personal innovativeness in the domain of IT (PIIT)*, adapted from Agarwal and Prasad (1998). Participants responded to each of these items on a 7-point Likert scale.

We recruited a total of 300 experiment participants from Amazon Mechanical Turk, with 103 participants randomly assigned to the control group, 96 to the early-error group, and 101 to the late-error group. As a randomization check, we applied a one-way ANOVA to test whether the participants' initial predictions for the first two rounds (i.e., before any treatment took place) were different across the three groups, and we found no significant difference ($F = 0.669,\ p = 0.414$). Similarly, we found no significant difference across groups in participants' choices to rely on the algorithm versus their initial predictions for the first two rounds ($F = 0.037,\ p = 0.848$). Finally, χ^2 tests showed no significant differences (all $p > 0.16$) in the distributions of gender, age, education, race, annual income, and country of origin across the three groups. Therefore, we conclude that the randomization was successful.

support to choosing the 12th round as the late-error treatment round.

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3.2 Analyses and Results

As model-free evidence of treatment effects, Figure 1 plots the percentage of participants who relied on the algorithm's prediction for each task. When compared with the control group, we observe a clear drop in the reliance on algorithm prediction immediately after the treatment rounds (i.e., round 3 for the early-error group and round 13 for the late-error group, respectively). The drop was especially sharp for participants in the early-error group, whose average reliance remained lower than that of the control group in all subsequent rounds. In contrast, despite an immediate reliance drop in round 13 for the late-error group, the average reliance quickly returned to a comparable level with the control group in the following rounds.

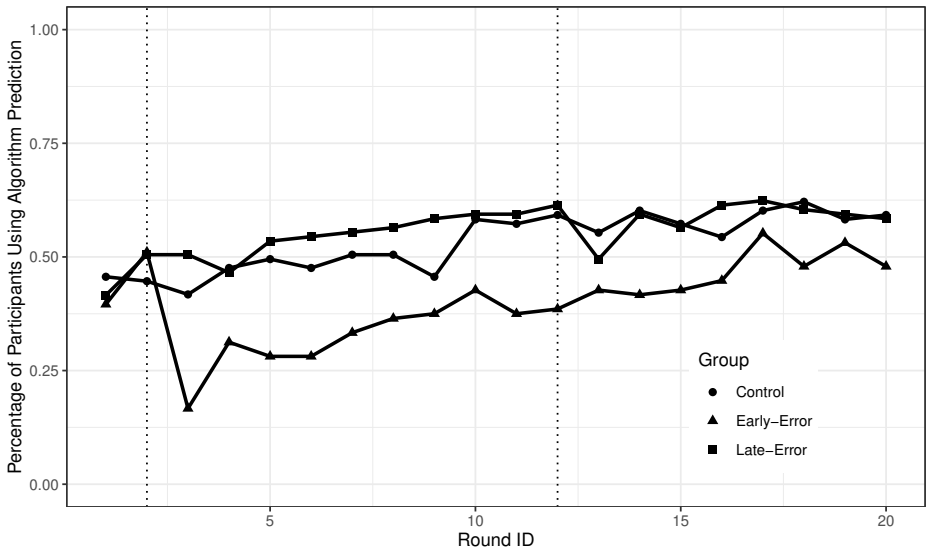


Figure 1: Percentage of participants using the algorithm predictions for each task. Vertical dotted lines mark the manipulated rounds for the two treatment groups respectively.

We next analyzed the treatment effects on algorithm reliance using regression methods. Leveraging the randomly assigned experiment conditions, we used a relative time difference-in-differences (DiD) logistic regression specifications.[5] Regression models (1) and (2) below estimate the treatment effects by comparing the control group with the early-error group

[5]The validity of a DiD regression depends on the "parallel trend assumption", requiring that the control group and the treatment group share the same trend of algorithm reliance prior to the treatment. This assumption is satisfied in our study because of successful randomization.

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and the late-error group, respectively.

$$logit(\Pr(Reliance_{it}=1)) = \beta_0 + \beta_1 Treat_i + \sum_{k=1}^{18} \alpha_k After_{kt} + \sum_{k=1}^{18} \gamma_k After_{kt} \cdot Treat_i + \boldsymbol{\Psi X_i} + \boldsymbol{\Theta Z_{it}} + \varepsilon_{it} \quad (1)$$

$$logit(\Pr(Reliance_{it}=1)) = \beta_0 + \beta_1 Treat_i + \sum_{k=1}^{8} \alpha_k After_{kt} + \sum_{k=1}^{8} \gamma_k After_{kt} \cdot Treat_i + \boldsymbol{\Psi X_i} + \boldsymbol{\Theta Z_{it}} + \varepsilon_{it} \quad (2)$$

The dependent variable, $Reliance_{it}$, is a binary indicator that takes value 1 if participant i chooses to rely on algorithm's prediction in round t, and takes value 0 otherwise. $Treat_i$ indicates whether participant i is in a treatment group or control group. In regression (1), $Treat_i = 1$ if i is assigned to the early-error group and $Treat_i = 0$ if i is assigned to the control group. In regression (2), $Treat_i = 1$ if i is assigned to the late-error group and $Treat_i = 0$ if i is assigned to the control group. $After_{kt}$ are a series of binary indicators of k rounds *after* the treatment round. Specifically in (1), for $k \in \{1, \ldots, 18\}$ (i.e., rounds 3 to 20), $After_{kt} = 1$ for all the rounds $t \geq (2 + k)$ and $After_{kt} = 0$ otherwise; in (2), for $k \in \{1, \ldots, 8\}$ (i.e., rounds 13 to 20), $After_{kt} = 1$ for all round $t \geq (12 + k)$ and 0 otherwise. The interaction terms between $Treat_i$ and $After_{kt}$ are also included, and their coefficients γ_k capture the treatment effects of all post treatment tasks.

We controlled for the characteristics of participants and prediction tasks. For participant-level controls (i.e., $\boldsymbol{X_i}$), we included six demographic variables including gender, age group, education level, race, annual income level, and country of residence. We also included two self-reported measures that capture participants' *perception* of their prediction accuracy, including their confidence and the self-estimated MAE of their initial predictions. To account for participants' *actual* abilities in making predictions, we controlled for the actual MAE associated with their initial predictions.[6] Additionally, we controlled for participants' self-reported trust and PIIT scores. We obtained a Cronbach's alpha of 0.85 for the three trust items, and 0.78 for the four PIIT items, indicating good internal validity. We used the average response values to measure the two constructs and included them in the model. At

[6]Note that we control for the MAE of participants' initial predictions rather than final predictions because the latter are affected by their choices of using algorithms.

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the prediction task level, we included two control variables (i.e., $\boldsymbol{Z_{it}}$). First, a participant's own prediction error in the previous round could affect her reliance on the algorithm in the current round. Therefore, at round t, we controlled for the absolute error of the participant's initial prediction in round $t-1$ (set to 0 when $t=1$). Second, we added the round index t as another control variable to account for changes in algorithm reliance as a participant got more familiar with the algorithm through repeated interactions.

We estimated regressions (1) and (2) as logistic regressions with cluster-robust standard errors (where each participant defines a cluster). Figure 2 shows the estimated γ_k (i.e., the treatment effects) and their 95% confidence intervals from the two regressions. The regression tables are omitted due to space limits.

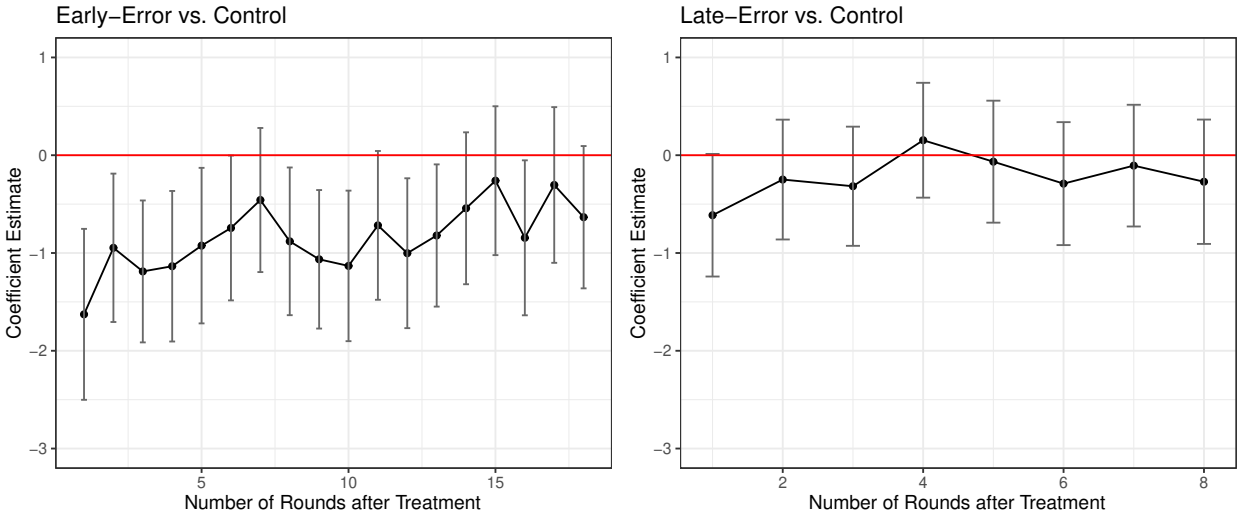


Figure 2: Study 1: Estimation of Treatment Effects and 95% Confidence Intervals for Decision to Rely on Algorithm's Predictions.

Several findings are worth noting. In the early-error group, we observed a strong decline in user reliance immediately after the substantial algorithm error was displayed (i.e., in the first round following the treatment). In other words, participants became much less likely to rely on the algorithm after seeing it making a big mistake. Also, this adverse treatment effect persists over multiple rounds after the treatment. Even though there were a few rounds with insignificant effects (at 5% level), we did not observe a stable pattern of multiple

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consecutive rounds with insignificant treatment effects. It indicates that participants exposed to a substantial error at the early stage of their interactions with the algorithm never fully restored their reliance on the algorithm during the rest of the experiment. In comparison, in the late-error group, except for a marginally significant (at 10% level) treatment effect in the immediate post-treatment round, we found insignificant effects in all subsequent rounds. This suggests that seeing a substantial error in the late stage of the experiment only temporarily affected users' reliance on the algorithm at first, and then the reliance level became virtually indistinguishable from that of the control group.

In summary, experiment results of Study 1 demonstrate that the impact of experiencing a substantial algorithm error on users' algorithm reliance is contingent on when the error occurs. If the error occurs early in the course of user-algorithm interactions, users' reliance will drop significantly and may not recover for a long time (if at all). It is likely that, when the error occurs early, participants form an inferior initial impression of the algorithm's competence, and such an impression is hard to change. In contrast, if the error occurs at a late stage when users have gained sufficient familiarity with the system's predictive performance, they are much more forgiving and their reliance barely decreases.

4 Study 2: Enhancing Reliance Recovery by Giving Control

In Study 2, we explored whether giving participants more control over how to use algorithm's predictions would affect the differential impact of early vs. late errors on participants' reliance on the algorithm.

4.1 Experiment Design

The procedure of Study 2 was similar to that of Study 1, except for one key change in how participants could use the algorithm's predictions in their decision making for each task.

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After a participant made an initial prediction for a task, we showed her the algorithm's prediction and gave her an option to adjust her initial prediction by entering a new value as her final prediction for the task. If she chose not to make any adjustment, her initial prediction was used as the final prediction. Compared to Study 1—where the participants decided between a complete delegation and not using the algorithm at all—the design of Study 2 reflected a more collaborative interaction between human and algorithm.[7]

We recruited a total of 302 participants for Study 2 from Amazon Mechanical Turk, with 105 participants randomly assigned to the control group, 95 to the early-error group, and 102 to the late-error group. None of the participants of Study 2 had previously enrolled in Study 1. To check randomization, we again applied a one-way ANOVA to compare the participants' predictions for the first two rounds across the experimental groups, and found no significant difference in their initial predictions ($F = 0.090,\ p = 0.764$) or their final predictions ($F = 0.013,\ p = 0.909$). We also found no significant differences in participants' demographics across groups. Overall, the randomization was successful.

4.2 Analyses and Results

We used two outcome measures as proxies for user's algorithm reliance. First, we examined *whether or not* a participant adjusted her prediction after seeing the algorithm's prediction. This serves as a binary measure of reliance. Second, we considered *how much* the participant adjusted her predictions, as a more granular, continuous measure of reliance.

4.2.1 Reliance Measure 1: Decision to Adjust Initial Predictions

We start by plotting the percentage of participants who adjusted their initial predictions after seeing the algorithm's predictions; see Figure 3. For both treatment groups, we observe

[7]Because the algorithm's prediction for each task was visible in this experiment, a potential concern is that participants may not make serious attempts to provide accurate initial predictions on their own, but rather just copy the algorithm's predictions. To check this possibility, we used two-sample t tests to compare the initial predictions between studies 1 and 2 on each task. After applying Bonferroni correction to multiple testing, the initial predictions were different only on 2 out of 20 tasks (at 5% significance level). Therefore, we conclude that participants of Study 2 did not systematically reduce the efforts to make initial predictions.

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a clear decline in user reliance in the round immediately following the treatment. However, within only a few rounds users' reliance quickly restored to a level comparable to the control group. This offers descriptive model-free evidence of reliance recovery. In addition, we observe that the highest number of participants adjusted their initial predictions in the treatment rounds (i.e., rounds 2 and 12 for the early-error and late-error groups, respectively), likely because the substantial algorithm error misled participants to believe that their initial predictions were far off. We removed data of these rounds in the following regression analyses, because it is not representative of participants' typical adjustment behaviors and would bias our results. Specifically, data from round 2 was removed for the early-error group and data from round 12 was removed for the late-error group.

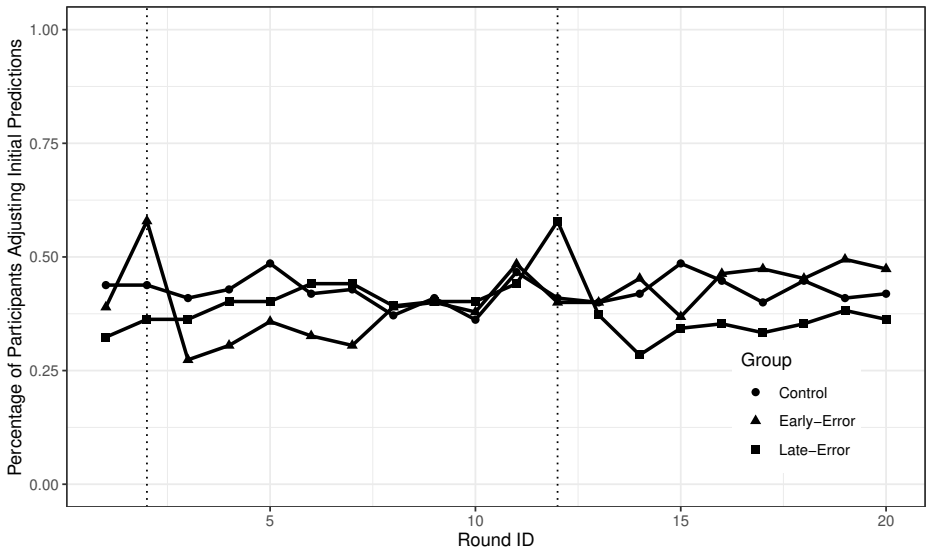


Figure 3: Percentage of participants adjusting initial predictions for each task. Vertical dotted lines mark the manipulated rounds for the two treatment groups respectively.

We next estimated the same logistic DiD regressions discussed in §3.2, to understand the treatment effects on participants' decisions to adjust predictions in light of algorithm predictions. The coefficients of treatment effects are plotted in Figure 4.

For the early-error group, we did not observe any significant reductions in the choices of making prediction adjustments throughout the experiment, even in the round immediately following the treatment. Similarly, for the late-error group, most of the treatment effects

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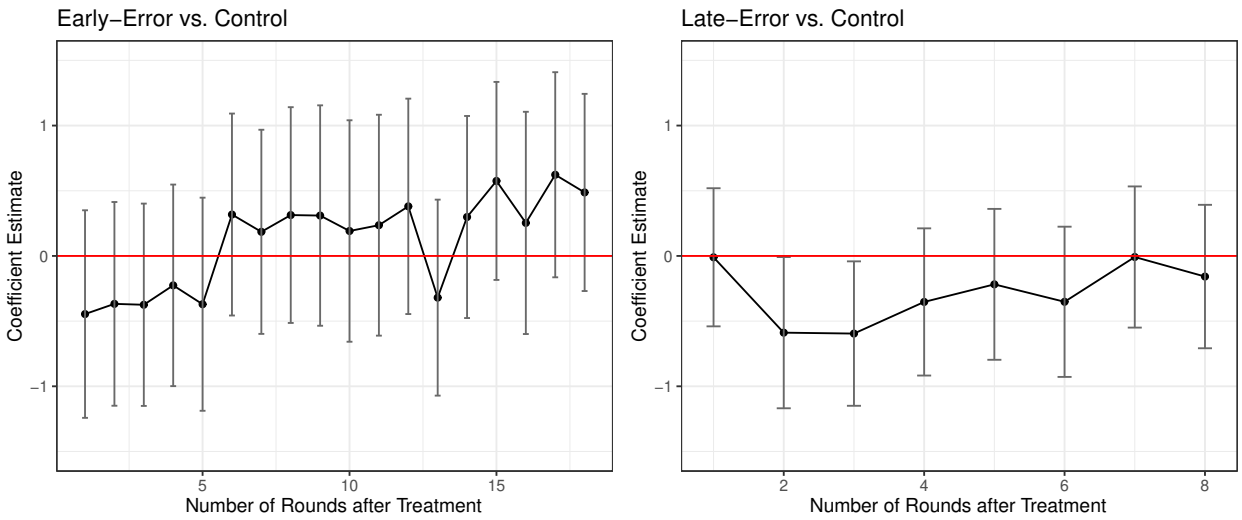


Figure 4: Study 2: Estimation of Treatment Effects and 95% Confidence Intervals for Decision to Adjust.

were insignificant, except in the 2nd and 3rd rounds after treatment, where we observed marginal significance at the 5% level. These two marginally significant effects were caused by a decline of reliance in the late-error group and an increase of reliance in the control group in these two rounds (see Figure 3 for visual evidence). Given that there was no significant effect even in the round that immediately followed the error round, it is unlikely that the two significant effects were due to our treatment. Overall, based on the binary measure of reliance, participants in both early-error and late-error groups did not significantly lower their reliance on the algorithm after observing a substantial error.

4.2.2 Reliance Measure 2: Degree of Adjustment

The second reliance measure examines the degree of adjustment made towards the algorithm's prediction. Specifically, for a participant i at round t, based on i's initial prediction ($InitPred_{it}$), the algorithm's prediction ($AlgoPred_{it}$), and i's adjusted prediction ($AdjPred_{it}$), we calculated the *weight* of the algorithm's prediction in the final prediction as $Weight_{it} = \left|\frac{AdjPred_{it} - InitPred_{it}}{AlgoPred_{it} - InitPred_{it}}\right|$. The weight equals 0 if $AdjPred_{it} = InitPred_{it}$ (i.e., no adjustment is made), and equals 1 if $AdjPred_{it} = AlgoPred_{it}$ (i.e., fully converts to algo-

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rithm's prediction). Thus, the weight intuitively captures the degree of user's reliance on the algorithm when constructing her final prediction.

For ease of interpretation, we excluded two types of data from our analyses. First, we removed instances where $AlgoPred_{it} = InitPred_{it}$ (i.e., user's initial prediction was the same as the algorithm's prediction), which would cause $Weight_{it}$ uncomputable. Second, we removed instances where $(AdjPred_{it} - InitPred_{it}) \times (AlgoPred_{it} - InitPred_{it}) < 0$, because they represent cases where, after observing the algorithm's prediction, the participant adjusted her prediction in the *opposite* direction.[8] The two types of instances accounted for 9.2% of the data. Additionally, it is also possible for $Weight_{it}$ to be larger than 1, which represents "over-adjustment." We included these instances (less than 4% of our data) in the analyses, although their exclusion would not change our findings.

Figure 5 illustrates the average weight of adjustment for each task across the three experimental groups. We observed similar patterns as in Figure 3: after the initial drops following the treatment rounds, the weight of adjustment quickly recouped to a comparable level with the control group within a few rounds. For both treatment groups, we again observed a high weight of adjustment during the respective treatment rounds, largely due to the presence of a substantial algorithm error. Same as before, data from those rounds were removed accordingly for the regression analyses.

We next ran the same DiD specifications, but as linear regressions instead of logistic regressions, to estimate the treatment effects. The DiD coefficients are plotted in Figure 6. The results showed that the treatment effects were largely insignificant for both early-error and late-error groups, offering no systematic evidence of reliance drop as a result of the substantial algorithm error. There were a few sporadic cases of significant effects. In the early-error group, the 15th and 17th rounds after treatment saw positive effects that were significant at 5% level. However, both were followed by a round with an insignificant treatment effect, and we therefore cannot conclude that the early-error group reached higher

[8] More specifically, it corresponds to two cases: (i) $AdjPred_{it} < InitPred_{it} < AlgoPred_{it}$ or (ii) $AlgoPred_{it} < InitPred_{it} < AdjPred_{it}$. The rationale for such adjustment behavior is not entirely clear.

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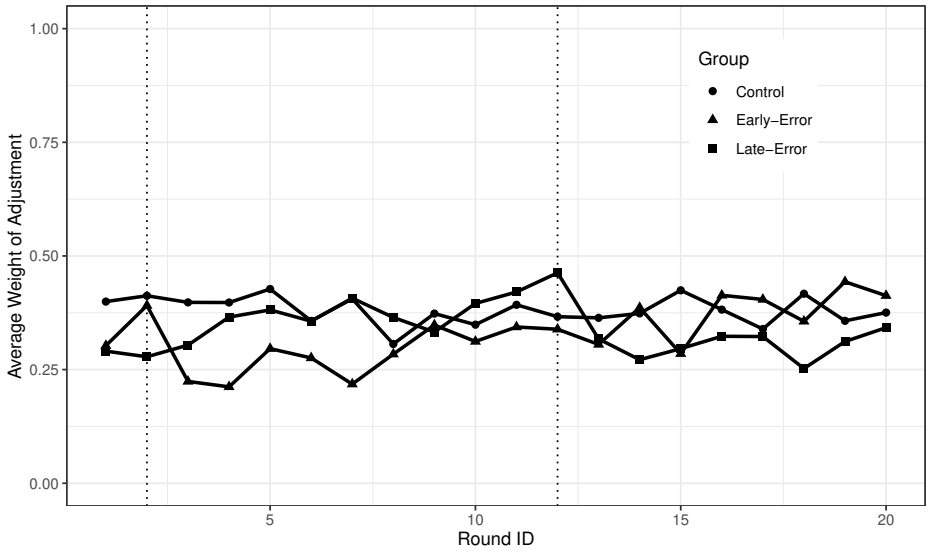


Figure 5: Average Weight of Adjustment for each task. Vertical dotted lines mark the manipulated rounds for the two treatment groups respectively.

reliance than the control group. In the late-error group, the 6th round after treatment had a negative effect, due to a sizable reliance drop in the late-error group and a reliance increase in the control group in that round (see Figure 5). Because none of the previous rounds had significant effects, this is more likely a one-off instance (e.g., due to noise in the data).

Overall, our experimental results of Study 2 demonstrated that when participants were given the ability to adjust their predictions based on the algorithm's predictions, their reliance on the algorithm was minimally affected by the exposure to a substantial algorithm error, even if the error occurred in the early stage. In other words, when people retain the control over the final predictions, their willingness to incorporate the algorithm's predictions in their decision making do not deteriorate even after the system shows a substantial error.

5 Discussion

In this work, we study how the timing of algorithmic errors influence users' reliance on algorithms. In our first study where users chose to either fully rely on their own predictions or completely delegate the decision to an algorithm, we found that algorithmic errors that

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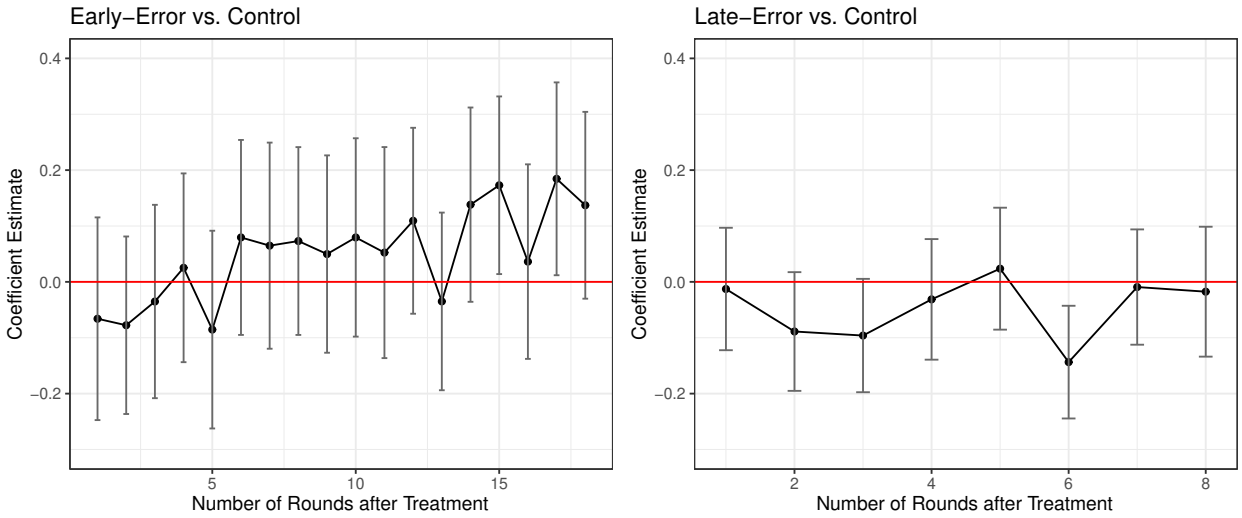


Figure 6: Study 2: Estimation of Treatment Effects and 95% Confidence Intervals for Weight of Adjustment.

occurred in the early stage were more detrimental to users' reliance than errors that occurred in the late stage. While the level of reliance decreased immediately after error occurrence, users who experienced errors in a late stage quickly restored their reliance, whereas the recovery was stagnant for users who experienced errors in an early stage. In fact, we did not observe a stable pattern of full recovery in users' reliance following the early error. In the second study, we further investigated a different context where human decision makers and algorithms interact in a more collaborative fashion. Our results show that, when users could view and adjust the algorithm's prediction before making a final decision, the adverse effect of substantial errors (both at an early stage or late stage) that we observed in the first study was mitigated. When users could retain full control over final decisions, they were willing to incorporate the algorithm's predictions into their decision making, even immediately after they witnessed a substantial algorithm error.

Our findings bear both theoretical importance and practical relevance. From a theoretical perspective, we contribute to existing work on algorithm aversion by showing that not all errors affect users' reliance to the same degree—it depends on when the errors occur and how the algorithm is being used. When an algorithm is fully delegated with the decisions,

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making a substantial error before users develop sufficient experience with it can destroy user reliance, whereas the same error occurring after sufficient interactions impose much smaller damage. However, when human users are in charge of decision making, their reliance on the algorithm becomes more robust against an early error. These theoretical insights can also inform the practical design of algorithmic tools. For example, when a system needs to explore decisions with a high likelihood of making substantial errors, it may be better to postpone the risky explorations until users have become familiar with the system and are therefore more forgiving of the errors. Furthermore, having human users in the loop can also mitigate aversion following algorithm errors.

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